

Final Year Project: smartRoute MaaS Platform

Work Division and Technical Responsibilities (CSE Team)

Member 1: Frontend & Visualization Engineer

Core Focus: User interface development, progressive web features, rendering interactive maps, and data analysis dashboards[cite: 3, 4].

- **Tech Stack:** React, HTML5/CSS3, JavaScript/TypeScript, Leaflet.js or Mapbox GL JS[cite: 4].
 - **Detailed Module Responsibilities:**
 - **Progressive Web App (PWA) Setup:** Configure custom service workers and web app manifests so the platform loads instantly, provides reliable navigation, and handles incoming search requests gracefully[cite: 4].
 - **Multimodal Map Visualizer:** Integrate a map framework to visually display integrated transport networks, routes, bus paths, and pedestrian tracks dynamically[cite: 3, 4].
 - **User Input Components:** Build interactive forms for commuters to input origin/destination locations, select dates/times, and filter by journey type (departure/arrival preferences)[cite: 1, 4].
 - **Mobility Analytics Dashboard:** Build an interactive portal filled with charts, geographic heatmaps, and graphs to help traffic managers visualize travel patterns, identify hotspots, and monitor network performance metrics[cite: 3, 4].
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Member 2: Backend & Routing Logic Engineer

Core Focus: Developing backend server infrastructure, routing computations, and traffic scheduling systems[cite: 4].

- **Tech Stack:** Python, Django or FastAPI, OpenTripPlanner (OTP) engine[cite: 4].

- **Detailed Module Responsibilities:**

- **Server Architecture:** Build a robust, scalable backend API in Python to process and deliver data between the React interface and the spatial database[cite: 4].
 - **OpenTripPlanner Configuration:** Set up, deploy, and manage the OpenTripPlanner engine to generate multi-modal travel options combining buses, trains, and walking networks[cite: 4].
 - **Algorithmic Pathfinding:** Implement Dijkstra's Algorithm within the server logic to execute rapid shortest-path network calculations across the city transit grid[cite: 4].
 - **Traffic Scheduling & Request Handling:** Structure a Round Robin (RR) scheduling mechanism to distribute search traffic seamlessly, ensuring optimal performance and server stability during high-load periods[cite: 4].
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Member 3: Database & GIS Data Manager

Core Focus: Handling spatial databases, data parsing workflows, and managing the core pipeline received from the Civil Engineering team[cite: 4].

- **Tech Stack:** PostgreSQL, PostGIS extension, Python data-cleaning scripts[cite: 4].
 - **Detailed Module Responsibilities:**
 - **Civil Team Technical Liaison:** Act as the primary technical point-of-contact to receive raw schedules, stop coordinates, and geometric transit paths from the Civil Engineering group[cite: 4].
 - **GTFS Data Pipeline:** Write data conversion scripts to standardize the Civil team's raw timetables into standard General Transit Feed Specification (GTFS) database feeds[cite: 4].
 - **Spatial Database Design:** Deploy a relational PostgreSQL database initialized with PostGIS extensions to store, index, and query geographic map shapes and coordinate sets[cite: 4].
 - **Query Optimization:** Code high-performance SQL scripts to execute rapid spatial joins on bus and transit lines, minimizing data lookup delays for the routing engine[cite: 2].
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Member 4: AI Systems & Validation Engineer

Core Focus: Implementing natural language travel tools, personalized recommendation modules, and coordinating final system verification testing[cite: 3, 4].

- **Tech Stack:** Python, Pandas, Gemini API or open-source LLM frameworks[cite: 4].

- **Detailed Module Responsibilities:**

- **AI Travel Assistant:** Code a conversational interface capable of parsing natural language travel queries to provide clear text-based route summaries and travel advice[cite: 3, 4].
- **Personalized Recommendation Engine:** Utilize Python and Pandas to analyze tracking statistics, compute multi-modal fare estimations, and provide customized trip suggestions based on historical user behavior[cite: 4].
- **Transfer Guidance Logic:** Program specialized backend logic to generate step-by-step guidance for complex route switchovers between independent transit modes[cite: 4].
- **Validation Framework:** Lead the final project evaluation by designing verification protocols to benchmark system response times, database query efficiency, and route accuracy using real city travel scenarios[cite: 3, 4].